2022 SSMO Tiebreaker Round

SMO Team

Tiebreaker Round Problem 1: For all positive integers n, let $S(n)$ denote the least positive integer x such that $n + x$ is a palindrome. Find the value of $\sum_{n=1}^{100} S(n)$.

Tiebreaker Round Problem 2: Let $P(x) = x^3 - 7x^2 + 9x - 2$. If $P(x) = (x - a)^3 + b(x - a) + c$ where a, b, c are real numbers, then the value of $a - b - c$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Tiebreaker Round Problem 3: Find the sum of the 5 smallest positive prime numbers p such that $3^n + 4^n \equiv 0 \pmod{p}$ has no positive integer solutions for n.